Algebraic Expressions

Solutions

Algebraic Expression Starter

Solution 2

1. Simplify

a) $4m^2n + 5mn^2 - 2m^2n + mn^2 - 3mn^2$

b) $3x^2 - 5x + 2 + 3x^2 - 7x - 12$

1. Simplify

a) $2m^2n + 3mn^2$ b) $6x^2 - 12x - 10$

2. Single power of 2

a) $2^5 \times 2^3$ b) $2^6 \div 2^2$ c) $(2^3)^2$

2. Single power of 2

a) 2^8 b) 2^4 c) 2^6

3. Expand

a) $3(x + 4)$ b) $5(2 - 3x)$ c) $6(2x - 5y)$

3. Expand

a) $3x + 12$ b) $10 - 15x$ c) $12x - 30y$

4. Highest common factor

a) $24, 16$ b) $6x, 8x^2$ c) $4xy^2, 3xy$

4. Highest common factor

a) 8 b) $2x$ c) xy

5. Simplify

a) $\frac{10x}{5}$ b) $\frac{20x}{2}$ c) $\frac{40x}{24}$

5. Simplify

a) $2x$ b) $10x$ c) $\frac{5x}{3}$

Chapter Overview

Solution 3

1: Basic Index Laws

2: Expand brackets

3: Factorise quadratics/cubics

4: Fractional/Negative Powers

5: Surds

1: Index Laws

Solution 4

In 3^5, 3 is the base and 5 is the exponent (or index).
The whole expression is a power.
Use precise language: when multiplying powers with the same base, add the indices.

$$a^m a^n = a^{m+n}$$
$$a^m / a^n = a^{m-n}$$
$$(a^m)^n = a^{mn}$$
$$(ab)^n = a^n b^n$$

1 Simplify $(a^3)^2 \times 2a^2$.

1
$a^6 \times 2a^2 = 2a^8$

2 Simplify $(4x^3y)^3$.

2
$64x^9y^3$

3 Simplify $2x^2(3+5x) - x(4-x^2)$.

3
$6x^2 + 10x^3 - 4x + x^3 = 11x^3 + 6x^2 - 4x$
Take care with the positive x^3 term.

4 Simplify $\frac{x^3-2x}{3x^2}$.

4
$\frac{x^3}{3x^2} - \frac{2x}{3x^2} = \frac{x}{3} - \frac{2}{3x}$
Split a fraction across terms in its numerator, never its denominator.

Test Your Understanding — Solution 5

1 Simplify

$$\left(\frac{2a^5}{a^2}\right)^2 \times 3a$$

1

$(2a^3)^2 \times 3a = 4a^6 \times 3a = 12a^7$

2 Simplify

$$\frac{2x + x^5}{4x^3}$$

2

$$\frac{1}{2x^2} + \frac{x^2}{4}$$

3 Expand and simplify

$2x(3 - x^2) - 4x^3(3 - x)$

3

$6x - 2x^3 - 12x^3 + 4x^4 = 4x^4 - 14x^3 + 6x$

4 Simplify

$2^x \times 3^x$

4

$(2 \times 3)^x = 6^x$

Use $(ab)^n = a^n b^n$ in reverse.

Exercise 1A – Pearson Pure Y1 page 4 — Solution 6

Exercise 2: expand and simplify. Exercise 3: simplify each fraction.

2(o)

$x(3x^2 - 2x + 5)$

2(o) $3x^3 - 2x^2 + 5x$

2(p)

$7y^2(2 - 5y + 3y^2)$

2(p) $14y^2 - 35y^3 + 21y^4$

2(q)

$-2y^2(5 - 7y + 3y^2)$

2(q) $-10y^2 + 14y^3 - 6y^4$

2(r)

$7(x - 2) + 3(x + 4) - 6(x - 2)$

2(r) $4x + 10$

2(s)

$5x - 3(4 - 2x) + 6$

2(s) $11x - 6$

2(t)

$3x^2 - x(3 - 4x) + 7$

2(t) $7x^2 - 3x + 7$

2(u)

$4x(x + 3) - 2x(3x - 7)$

2(u)

$26x - 2x^2$

2(v)

$3x^2(2x + 1) - 5x^2(3x - 4)$

2(v)

$23x^2 - 9x^3$

3(d)

$$\frac{8x^3 + 5x}{2x}$$

3(d) $4x^2 + \frac{5}{2}$

3(e)

$$\frac{7x^7 + 5x^2}{5x}$$

3(e) $\frac{7}{5}x^6 + x$

3(f)

$$\frac{9x^5 - 5x^3}{3x}$$

3(f) $3x^4 - \frac{5}{3}x^2$

Exercise 1A – Pearson Pure Y1 page 4 — Solution 7

1. MAT 2006 1A Which number is largest?

A) $((2^3)^2)^3$ B) $(2^3)^{2^3}$

C) $2^{(3^2)^3}$ D) $2^{3^{2^3}}$

1. MAT 2006 1A

A) 2^{18}, B) 2^{24}, C) 2^{729}, D) 2^{6561}

D is largest.

2. MAT 2012 1B Let $N = 2^k 4^m 8^n$, where k, m, n are positive integers. When is N definitely a square?

A) k even B) $k+n$ odd

C) k odd and $m+n$ even D) $k+n$ even

2. MAT 2012 1B

$N = 2^{k+2m+3n}$

The exponent is even exactly when $k+n$ is even. Answer: D.

2: Expanding Brackets — Solution 8

Multiply each term in the first bracket by every term in the second bracket. Work systematically.

1

$(x-y)(x+y-1)$

1

$$x^2 + xy - x - xy - y^2 + y$$
$$= x^2 - y^2 - x + y$$

2

$(x+1)(x+2)(x+3)$

2

$$(x+1)(x^2+5x+6)$$
$$= x^3 + 5x^2 + 6x + x^2 + 5x + 6$$
$$= x^3 + 6x^2 + 11x + 6$$

For three or more brackets, expand two at a time.

Test Your Understanding

Solution 9

1 Expand $(x+5)(x-2)(x+1)$.

1

$$\begin{aligned}&(x+5)(x^2-x-2)\\ =\ &x^3-x^2-2x+5x^2-5x-10\\ =\ &x^3+4x^2-7x-10\end{aligned}$$

2 Expand $2(x-3)(x-4)$.

2

$$2(x^2-7x+12)=2x^2-14x+24$$

3 Expand $(2x-1)^3$.

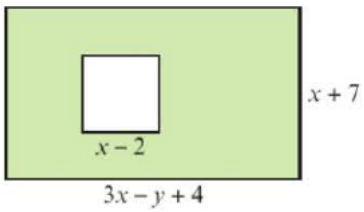
3

$$\begin{aligned}&(2x-1)(4x^2-4x+1)\\ =\ &8x^3-8x^2+2x-4x^2+4x-1\\ =\ &8x^3-12x^2+6x-1\end{aligned}$$

Exercise 1B - Pure Year 1/AS Page 6

Solution 10

3 The diagram shows a rectangle with a square cut out. The rectangle has length $3x-y+4$ and width $x+7$. The square has side $x-2$. Find an expanded and simplified expression for the shaded area.



3

$$\begin{aligned}\text{Shaded area}&=(x+7)(3x-y+4)-(x-2)^2\\ &=3x^2-xy+25x-7y+28\\ &\quad-(x^2-4x+4)\\ &=2x^2-xy+29x-7y+24\end{aligned}$$

5 Find a, b, c, d when

$(2x+5y)(3x-y)(2x+y)=ax^3+bx^2y+cxy^2+dy^3$

5

$$\begin{aligned}&(6x^2+13xy-5y^2)(2x+y)\\ &=12x^3+32x^2y+3xy^2-5y^3\end{aligned}$$

$$a=12,\quad b=32,\quad c=3,\quad d=-5$$

1. MAT 2002 1B Which identities are correct?

(i) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(x-z)(y-x)$

(ii) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(z-x)(y-x)$

(iii) $yz(x+y)+zx(z+x)+xy(y+x)$
$=(z+y)(z+x)(y+x)$

1. MAT 2002 1B

Only (ii) is an identity.

2. MAT 2007 1E For integers x, n, consider
$(1-x)^n(2-x)^{2n}(3-x)^{3n}(4-x)^{4n}(5-x)^{5n}$
Which statement holds?

A. Negative when $n>5$ and $x<5$.

B. Negative when n is odd and $x>5$.

C. Negative when n is a multiple of 3 and $x>5$.

D. Negative when n is even and $x<5$.

2. MAT 2007 1E

B. If $x>5$, all five bases are negative. The combined sign is $(-1)^{15n}$, which is negative when n is odd.

3: Factorising

Solution 12

Factorising reverses expanding: write an expression as a product of factors.

$x(x+1)(x+2)$

Factorised: a product of three linear factors. A linear expression has the form $ax+b$.

$x(x+1)+(x-1)(x+1)$

Not yet factorised: the outermost operation is addition.

Basic examples

1 $x^3+x^2=$ $x^2(x+1)$

2 $4x-8xy=$ $4x(1-2y)$

Factorising Quadratics

Solution 13

Recap

For $x^2 + bx + c$, find two numbers that add to b and multiply to c. The coefficient is the number multiplying a term.

1 $x^2 - 5x - 14 =$

$(x - 7)(x + 2)$

When the coefficient of x^2 is not 1

Guess and check the brackets, or split the middle term.

2 $2x^2 + 5x - 12 =$

$(2x - 3)(x + 4)$

Split the middle term Find two numbers with sum 5 and product $2(-12) = -24$.

Split the middle term $8 + (-3) = 5, \qquad 8(-3) = -24$

Working Use those numbers to split $5x$, then factorise both pairs.

Working

$$\begin{aligned} 2x^2 + 5x - 12 &= 2x^2 + 8x - 3x - 12 \\ &= 2x(x + 4) - 3(x + 4) \\ &= (2x - 3)(x + 4) \end{aligned}$$

Other Factorisations

Solution 14

Difference of two squares

$a^2 - b^2 = (a + b)(a - b)$. Always look for a common factor first.

1 $4x^2 - 9 =$

$(2x + 3)(2x - 3)$

2 $x^3 - x =$

$$\begin{aligned} & x(x^2 - 1) \\ &= x(x + 1)(x - 1) \end{aligned}$$

3 $x^3 + 3x^2 + 2x =$

$$\begin{aligned} & x(x^2 + 3x + 2) \\ &= x(x + 1)(x + 2) \end{aligned}$$

Test Your Understanding — Solution 15

1. Factorise completely

$6x^2 + x - 2$

1. Factorise completely

$$6x^2 + 4x - 3x - 2 = 2x(3x + 2) - (3x + 2)$$
$$= (3x + 2)(2x - 1)$$

2. Factorise completely

$x^3 - 7x^2 + 12x$

2. Factorise completely

$$x(x^2 - 7x + 12) = x(x - 3)(x - 4)$$

3. Challenge Factorise

$x^4 - 1$

3. Challenge

$$(x^2 + 1)(x^2 - 1) = (x^2 + 1)(x + 1)(x - 1)$$

4. Challenge Factorise

$x^3 - 1$

4. Challenge

$(x - 1)(x^2 + x + 1)$

Difference of cubes:
$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. This extension is beyond the usual A-level factorisation requirement.

Exercise 1C - Pure Y1 Page 8 — Solution 16

4 Factorise $x^4 - y^4$.

4

$$(x^2 + y^2)(x^2 - y^2) = (x^2 + y^2)(x + y)(x - y)$$

5 Factorise $6x^3 + 7x^2 - 5x$.

5

$$x(6x^2 + 7x - 5)$$
$$= x(6x^2 + 10x - 3x - 5)$$
$$= x(3x + 5)(2x - 1)$$

Challenge Write $4x^4 - 13x^2 + 9$ as four linear factors.

Challenge

$$(x^2 - 1)(4x^2 - 9)$$
$$= (x + 1)(x - 1)(2x + 3)(2x - 3)$$

4: Negative and Fractional Indices

Solution 17

$a^0 = 1, \quad a^{1/m} = \sqrt[m]{a}, \quad a^{n/m} = (\sqrt[m]{a})^n, \quad a^{-m} = \dfrac{1}{a^m}$

Use positive a for these fractional-index laws.

$\sqrt{9} = 3$, the positive square root; $\pm\sqrt{9}$ gives both 3 and -3.

1 Explain why $x^{1/2} = \sqrt{x}$ for $x > 0$.

1

$x^{1/2}x^{1/2} = x^1 = x$

Both sides denote the positive square root.

2 Evaluate $27^{-1/3}$.

2

$\dfrac{1}{\sqrt[3]{27}} = \dfrac{1}{3}$

3 Evaluate $32^{2/5}$.

3

$(\sqrt[5]{32})^2 = 2^2 = 4$

4 Simplify $(\frac{1}{9}x^6y)^{1/2}$, assuming $x, y > 0$.

4

$\dfrac{1}{3}x^3y^{1/2}$

5 Evaluate $(\frac{27}{8})^{-2/3}$.

5

$(\dfrac{8}{27})^{2/3} = (\dfrac{2}{3})^2 = \dfrac{4}{9}$

6 If $b = \frac{1}{9}a^2$, express $3b^{-2}$ in the form ka^n.

6

$3(\frac{1}{9}a^2)^{-2} = 3 \times 81a^{-4} = 243a^{-4}$

Writing a surd using indices

Solution 18

Example If $9\sqrt{3} = 3^k$, find k.

Example

Write everything with base 3:

$9\sqrt{3} = 3^2 \times 3^{1/2} = 3^{5/2} \quad \Rightarrow \quad k = \dfrac{5}{2}$

Use an improper fraction rather than a mixed number in an exponent.

Further Examples

Solution 19

1. Edexcel IAL C12 Jan 2019 Q2(c) Given $y = 2^x$, express $1/4^{2x-3}$ in terms of y.

1. Edexcel IAL C12 Jan 2019 Q2(c)

$$\frac{1}{4^{2x-3}} = 4^{-2x+3} = 2^{-4x+6}$$
$$= 2^{-4x}2^6 = 64(2^x)^{-4}$$
$$= \frac{64}{y^4}$$

2. Edexcel IGCSE May 2014 4H Given $(2^{1/2})^n = 2^x/8^y$, express n in terms of x, y.

2. Edexcel IGCSE May 2014 4H

$$2^{n/2} = \frac{2^x}{2^{3y}} = 2^{x-3y}$$
$$\frac{n}{2} = x - 3y$$
$$n = 2x - 6y$$

Write 8 as 2^3 to obtain a common base.

Test Your Understanding

Solution 20

Question Given $2^x 4^y = 1/(2\sqrt{2})$, express y as a function of x. (3 marks)

Question

$$2^x 2^{2y} = 2^{-3/2}$$
$$x + 2y = -\frac{3}{2}$$
$$y = -\frac{1}{2}x - \frac{3}{4}$$

Exercise 1D

Solution 21

4. Evaluate or simplify

4(a) $81^{1/4} =$

4(a) $\sqrt[4]{81} = 3$

4(b) $x(2x^{-1/3})^4 =$

4(b) $16x^{1-4/3} = 16x^{-1/3}$

5. Given $y = \frac{1}{8}x^3$, express in terms of x.

5(a) $y^{1/3} =$

5(a)
$$\left(\tfrac{1}{8}x^3\right)^{1/3} = \frac{x}{2}$$

5(b) $\frac{1}{2}y^{-2} =$

5(b)
$$\frac{1}{2}\left(\tfrac{1}{8}x^3\right)^{-2} = \frac{32}{x^6}$$

Challenge · MAT 2007 1A For integers r, s, when is this expression an integer?

$$\frac{6^{r+s}12^{r-s}}{8^r 9^{r+2s}}$$

Choose: $r + s \le 0$, $s \le 0$, $r \le 0$, or $r \ge s$.

Challenge · MAT 2007 1A

$$\frac{2^{3r-s}3^{2r}}{2^{3r}3^{2r+4s}} = 2^{-s}3^{-4s} = 162^{-s}$$

This is an integer precisely when $s \le 0$.

5: Surds

Solution 22

A surd is a root that does not simplify to a rational number. A rational number can be written a/b, where a, b are integers and $b \neq 0$.

$\sqrt{a}\sqrt{b} = \sqrt{ab}$, $\quad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}} \quad (a \ge 0, b > 0)$

1 $\sqrt{3} \times 2 =$

$2\sqrt{3}$

2 $3\sqrt{5} \times 2\sqrt{5} =$

30

3 $\sqrt{8} =$

$\sqrt{4}\sqrt{2} = 2\sqrt{2}$

4 $\sqrt{12} + \sqrt{27} =$

$2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$

5 $(\sqrt{8} + 1)(\sqrt{2} - 3) =$

$4 - 6\sqrt{2} + \sqrt{2} - 3$
$= 1 - 5\sqrt{2}$

Exercise 1E – Pure Y1 Page 13 Solution 23

3 Simplify $\sqrt{75} - \sqrt{12}$ in the form $a\sqrt{3}$, with a an integer.

3

$\sqrt{75} - \sqrt{12} = 5\sqrt{3} - 2\sqrt{3} = 3\sqrt{3}$

Extension

1. SMC 2014 Q24 Which is smallest?

$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$;
$9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$

1. SMC 2014 Q24

Each equals $\sqrt{n+1} - \sqrt{n}$. Rationalising gives $1/(\sqrt{n+1} + \sqrt{n})$, which decreases as n increases.

The largest n is 99, so the smallest is $10 - 3\sqrt{11}$.

2. SMC 2012 Q21 Which has no square root of the form $x + y\sqrt{2}$, with positive integers x, y?

$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$;
$54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$

2. SMC 2012 Q21

Squaring gives $x^2 + 2y^2 + 2xy\sqrt{2}$, so $xy = 6$.

The pairs $(1, 6), (2, 3), (3, 2), (6, 1)$ give constants $73, 22, 17, 38$.

Therefore the answer is $54 + 12\sqrt{2}$.

6: Rationalising The Denominator Solution 24

What can we multiply this surd by to obtain a rational number?

$\sqrt{5}\times$ Multiplier 1: $\sqrt{5}$ $=$ Result 1: 5

Rationalising a denominator makes it rational. Multiply the numerator and denominator by the same non-zero expression, so the value stays unchanged.

$\frac{1}{\sqrt{2}}\times$ Multiplier 2: $\frac{\sqrt{2}}{\sqrt{2}}$ $=$ Result 2: $\frac{\sqrt{2}}{2}$

This gives a simpler denominator and makes expressions involving surds easier to combine.

Examples

Solution 25

Worked examples · Rationalise the denominator

1 $\dfrac{3}{\sqrt{2}} =$

$\dfrac{3\sqrt{2}}{2}$

2 $\dfrac{6}{\sqrt{3}} =$

$\dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$

3 $\dfrac{7}{\sqrt{7}} =$

$\dfrac{7\sqrt{7}}{7} = \sqrt{7}$

4 $\dfrac{15}{\sqrt{5}} + \sqrt{5} =$

$3\sqrt{5} + \sqrt{5} = 4\sqrt{5}$

Test your understanding

5

$\dfrac{12}{\sqrt{3}}$

5

$4\sqrt{3}$

6

$\dfrac{2}{\sqrt{6}}$

6

$\dfrac{\sqrt{6}}{3}$

7

$\dfrac{4\sqrt{2}}{\sqrt{8}}$

7

2

More Complex Denominators

Solution 26

For a denominator with two terms, multiply by its conjugate: reverse the sign between the terms.

$\dfrac{1}{\sqrt{2}+1} \times$

1. Conjugate

$\dfrac{\sqrt{2}-1}{\sqrt{2}-1}$

$=$

2. Simplify

$\dfrac{\sqrt{2}-1}{2-1} = \sqrt{2}-1$

Why this works

$(a+b)(a-b) = a^2 - b^2$

The difference of two squares removes the surd from the denominator.

More Examples

Solution 27

1 Rationalise

$$\frac{3}{\sqrt{6}-2}$$

1

$$\frac{3(\sqrt{6}+2)}{(\sqrt{6}-2)(\sqrt{6}+2)} = \frac{3\sqrt{6}+6}{6-4} = \frac{3\sqrt{6}+6}{2}$$

2 Rationalise

$$\frac{4}{\sqrt{3}+1}$$

2

$$\frac{4(\sqrt{3}-1)}{3-1} = 2\sqrt{3}-2$$

3 Rationalise

$$\frac{3\sqrt{2}+4}{5\sqrt{2}-7}$$

3

$$\frac{(3\sqrt{2}+4)(5\sqrt{2}+7)}{50-49} = 30+21\sqrt{2}+20\sqrt{2}+28 = 58+41\sqrt{2}$$

Remember $(a-b)(a+b) = a^2-b^2$: use the difference of two squares to simplify each denominator.

Test Your Understanding

Solution 28

1. AQA IGCSE FM June 2013 Paper 1 Solve $y(\sqrt{3}-1) = 8$. Give y in the form $a+b\sqrt{3}$.

1. AQA IGCSE FM June 2013 Paper 1

$$y = \frac{8(\sqrt{3}+1)}{3-1} = 4+4\sqrt{3}$$

2 Rationalise and simplify $\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$.

2

$$\frac{(2\sqrt{3}-1)(3\sqrt{3}-1)}{27-1} = \frac{19-5\sqrt{3}}{26}$$

3 Rationalise and simplify $\frac{4}{\sqrt{5}-2}$.

3

$$\frac{4(\sqrt{5}+2)}{5-4} = 8+4\sqrt{5}$$

Writing surd expressions in power form

Solution 29

Writing surds in power form is useful in later calculus questions.

Example Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Example

$$\frac{4+4\sqrt{x}+x}{x} = \frac{4+4x^{1/2}+x}{x}$$
$$= 4x^{-1}+4x^{-1/2}+1$$

Expand, replace roots with powers, then split the fraction.

Test your understanding Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

Test your understanding

$$\frac{x^2+2x^{3/2}+x}{2x^{1/2}} = \frac{1}{2}x^{3/2}+x+\frac{1}{2}x^{1/2}$$

Exercise 1F (Page 15)

Solution 30

1. Rationalise and simplify

1(a) $\frac{1}{\sqrt{5}+2} =$

1(a) $\sqrt{5}-2$

1(b) $\frac{\sqrt{3}}{\sqrt{3}-1} =$

1(b) $\frac{3+\sqrt{3}}{2}$

1(c) $\frac{\sqrt{5}+1}{\sqrt{5}-2} =$

1(c) $7+3\sqrt{5}$

1(d) $\frac{2\sqrt{3}-1}{3\sqrt{3}+4} =$

1(d) $2-\sqrt{3}$

1(e) $\frac{5\sqrt{5}-2}{2\sqrt{5}-3} =$

1(e) $4+\sqrt{5}$

2 Expand $(\sqrt{5}+3)(\sqrt{5}-2)(\sqrt{5}+1)$.

2 4

3 Rationalise $\frac{3\sqrt{3}+7}{3\sqrt{3}-5}$ in the form $a+b\sqrt{3}$.

3 $31+18\sqrt{3}$

4 Solve $x(4-\sqrt{6})=10$.

4 $x = \frac{10}{4-\sqrt{6}} = 4+\sqrt{6}$

5 Solve $y(1+\sqrt{2})-\sqrt{2}=3$.

5 $y = \frac{3+\sqrt{2}}{1+\sqrt{2}} = 2\sqrt{2}-1$

6 Simplify $\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}$.

6 $2a+1-2\sqrt{a}\sqrt{a+1}$

Challenge Solve $\frac{\sqrt[4]{9}}{\sqrt[5]{27}} = \sqrt[x]{3}$, interpreting the right side as $3^{1/x}$.

Challenge

$$\frac{3^{2/4}}{3^{3/5}} = 3^{1/2-3/5} = 3^{-1/10}$$

$$\frac{1}{x} = -\frac{1}{10} \quad \Rightarrow \quad x = -10$$